UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education Ordinary Level

MARK SCHEME for the November 2005 question paper

5054 PHYSICS

5054/04 Paper 4, maximum mark 30

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1 **(a)** current increases/pointer moves to right
resistance decreases **[2]**

(b) scale/units marked on the meter **[1]**

[Total 3]

2 **(a)** **(i)** to enable a range of current/distance/light intensity to be measured **[1]**

(ii) 4 mA/5 mA/10 mA **[1]**

(b) table with headings: quantities distance, current (allow d, I)
units cm, mA
pairs of measurements entered in table **[3]**

(c) axes: correct way round, labelled quantity and unit
scales: more than ½ page, sensible (origin not needed)
6 points plotted accurately and neatly ± ½ square e.c.f. **(b)**
best fit line drawn, smooth curve, neatly **[4]**

(d) increasing distance decreases current/inverse relationship **[1]**

(e) shade around bulb/bulb is hot/filament is inside bulb/
uncertainty in position of LDR or ruler/ruler or circuit unsupported **[1]**

(f) lamp not over base/far from stand/could topple/is unstable **[1]**

[Total 12]

3 **(a)** **(i)** diagram of convex lens + Sun/distant object/parallel rays to lens
+ screen or rays to focus + f marked or ruler shown **[1]**

(ii) value for f too large or larger/image is further from lens **[1]**

(b) **(i)** rays hit mirror at 90°/reflected by mirror/
rays go back along the same path (can be shown on diagram) **[1]**

(ii) 1 no change/image in focus/image is dimmer **[1]**
2 image becomes blurred/out of focus/disappears **[1]**

[Total 5]

4 **(a)** cross-sectional area of the tube is constant/V = Axl **[1]**

(b) gas may have heated up/to allow the gas to cool down/
liquid or apparatus settles/so temperature remains constant **[1]**

(c) P x I calculated for both values/P doubles and I halves **[1]**

(d) to avoid/reduce (parallax) error/more accurate **[1]**

[Total 4]

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5 **(a)** brass/metal is a good conductor/heats up quickly/has high melting point/ does not melt/does not rust **[1]**

(b) temperature of water
initial and final temperatures/temperature rise of the water **[2]**

(c) any two from: cube may not all be at the same temperature/in the flame
time taken/heat lost by cube during transfer
some water may boil/spit out of beaker
beaker is also heated
human error in specified measurement
other sensible suggestion **[2]**

(d) valid comment linked to response in **(d)** **[1]**

[Total 6]
[Paper total 30]

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